

Same-well imaging and transcriptomics in a 50,000-well one-bead-one-compound microchip recovers compound-resolved phenotypes

Zafrens, Inc.

Abstract

Phenotypic screens trade throughput against biological depth: high-content imaging scales to millions of wells with a limited readout, while single-cell transcriptomics provides a deeper readout at substantially higher cost per compound. The Z-Screen platform addresses both axes through one-bead-one-compound combinatorial libraries profiled in 50,000-well microchips by paired imaging and mRNA sequencing. Each micro-well contains 5 to 10 cells, providing partial averaging over cellular heterogeneity while pairing both readouts on the same multicellular colony rather than on replicate populations exposed to the same treatment. ActiveSeq is the matched-readout layer of that platform.

Here we evaluate whether the two same-well readouts outperform either readout alone on a held-out compound classification task.

The analysis used a canonical ActiveSeq dataset of 20,222 matched wells across two pilot experiments (ACT010 and ACT011): 11,511 wells from a 35-compound control panel, and 8,711 ZEL026 dummy-bead wells retained as a negative-control QC stress test. After task-aware calibration of the per-well image embedding, both readouts were highly reproducible. Within-run split-half top-1 retrieval reached 98.8% in RNA and 94.2% in the calibrated image latent for ACT010, and 80.1% / 86.8% for ACT011, against a 2.9% top-1 chance level (35 classes). Compound centroids learned in ACT010 retrieved the same compounds in ACT011 at 71.4% top-1 (94.3% top-3) in RNA and 85.7% top-1 (100% top-3) in the calibrated image latent.

Same-well multimodality was strongly additive. A linear discriminant trained on merged image-plus-RNA features reached 55.5% balanced accuracy on held-out wells, against 41.4% for RNA alone and 29.2% for image alone. The ordering held within each run (ACT010: 60.8% merged vs 47.3% RNA; ACT011: 45.1% merged vs 30.5% RNA), and merged features improved per-compound recall for 35 of 35 compounds in the pooled analysis. The two views also preserved a shared map of compound relationships, with Spearman 0.718 (ACT010) and 0.607 (ACT011) between pairwise compound-distance matrices in image and RNA space, and per-run effect-size rankings transferred across runs at Spearman 0.799 (image) and 0.662 (RNA).

The closest public benchmark, scGeneScope, is a treatment-matched Cell Painting and scRNA-seq

dataset across 28 treatments in U2-OS cells with approximately 716,767 single-cell images and 627,704 single-cell RNA profiles. The present pilot is an order of magnitude smaller in total cells profiled (Z-Screen reads each well as a pseudobulk of 5 to 10 cells rather than at single-cell resolution) and uses a simpler imaging modality. Once task difficulty is normalized, the multimodal numbers fall in the same range: pooled merged features reach 19.4 times chance, against 18.0 times for the scGeneScope multimodal multiprofile reference; ACT010 alone reaches 21.3 times chance. Centered pseudobulk gene signatures retrieve the correct compound across runs at 48.6% top-1 and 68.6% top-3, with reproducible compound-specific gene programs. Together, these results establish that ActiveSeq supports same-well multimodal profiling, that calibration recovers compound-resolved structure not directly accessible in the original frozen image embedding, and that same-well fusion produces an additive gain on held-out classification.

Significance

Treatment-matched Cell Painting and Perturb-seq experiments yield two complementary descriptions of cell state on separate cell populations exposed to the same compound. ActiveSeq pairs both descriptions on the same multicellular colony in the same micro-well, on a chip designed for 50,000 wells in parallel. The present analysis demonstrates that this tighter pairing produces an additive predictive gain on held-out compound classification across all 35 compounds in two pilot experiments. For early-stage discovery, this shifts the readout from ranking wells to characterizing the cell-state response that distinguishes a hit.

Introduction

Early-stage discovery requires both throughput sufficient to cover relevant chemical space and biological resolution sufficient to characterize hit responses. Existing platforms typically optimize one axis at the expense of the other.

Cell Painting and the imaging-only ML platforms built on it (Recursion, Insitro, the Broad’s morphological profiling program) have driven imaging throughput up and per-image cost down to the point where multi-million-compound runs are routine [1]. The readout is comparatively shallow: a small set of fluorescent stains and the morphological features extractable from them by deep networks. Single-cell transcriptomics, through Perturb-seq [3] and platforms such as Tahoe-100M and scPerturb, provides a substantially deeper readout of cell state at higher cost per profile and lower compound concurrency. Public efforts to bridge the two have produced treatment-matched datasets, of which scGeneScope (a treatment-matched Cell Painting and scRNA-seq benchmark in U2-OS cells) is the strongest current reference [4]. Treatment-matched datasets pair imaging and sequencing across separate populations of cells exposed to the same compound; alignment is at the treatment level, not the cellular level.

Z-Screen occupies a different design point. Compounds are delivered through one-bead-one-compound combinatorial libraries and profiled in 50,000-well microchips by paired imaging and mRNA sequencing, at 5 to 10 cells per micro-well. The readout averages over limited cellular heterogeneity without incurring the per-profile cost of single-cell resolution. ActiveSeq is the matched-readout layer: each

per-well image is paired with a transcriptomic measurement from the same micro-well. The pairing is stricter than in treatment-matched public benchmarks because both readouts describe the same multicellular colony rather than two replicate populations. With same-well pairing established, the central evaluation question is whether combined image and RNA features outperform either modality alone on a held-out classification task. That is the question addressed here.

The analysis is structured around four questions: 1. Are matched compound measurements reproducible within a run? 2. Do image and RNA preserve a shared map of compound relationships? 3. Does same-well fusion produce a held-out classification gain over either single view? 4. Do low-pass gene programs remain interpretable enough to support downstream triage?

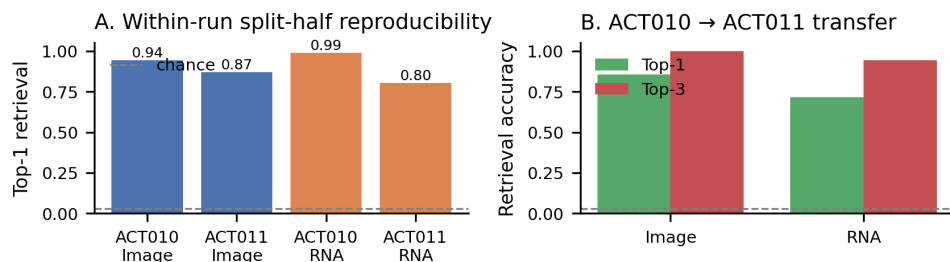
Results are then anchored against scGeneScope so the comparison can be read at the appropriate level of abstraction.

Results

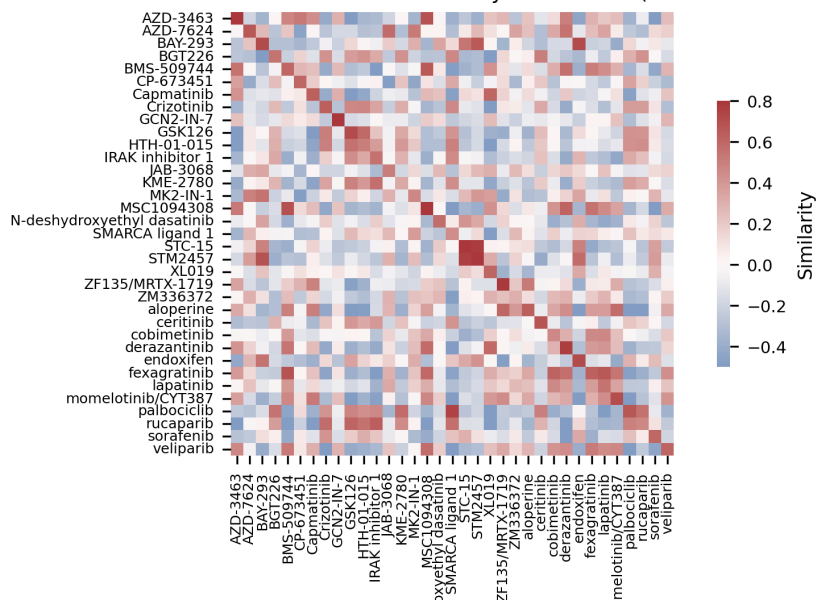
The 35-compound control panel is reproducible within run and transferable across runs

Wells for each control compound were split into two pseudo-replicates, and one half was queried for retrieval of the other among the 35 compounds. RNA latent space was highly stable: split-half top-1 retrieval was 98.8% in ACT010 and 80.1% in ACT011. The calibrated image latent was similar at 94.2% (ACT010) and 86.8% (ACT011). Top-3 retrieval reached 100% for RNA in ACT010 and 99.6% for the calibrated image latent.

Cross-run transfer is the more demanding criterion: a control-compound strategy is only useful if it transfers across experiments. Compound centroids learned in ACT010 retrieved the same compounds in ACT011 at 71.4% top-1 (94.3% top-3) in RNA and 85.7% top-1 (100% top-3) in the calibrated image latent, against a 2.9% top-1 chance level. Both readouts therefore support a reusable control panel across experiments. The image-side numbers also indicate that compound-identity information was largely present in the original embedding but obscured by the raw geometry; task-aware calibration made it accessible.



C. RNA centroid similarity across runs (ACT010 rows × ACT011 cols)



D. Image centroid similarity across runs

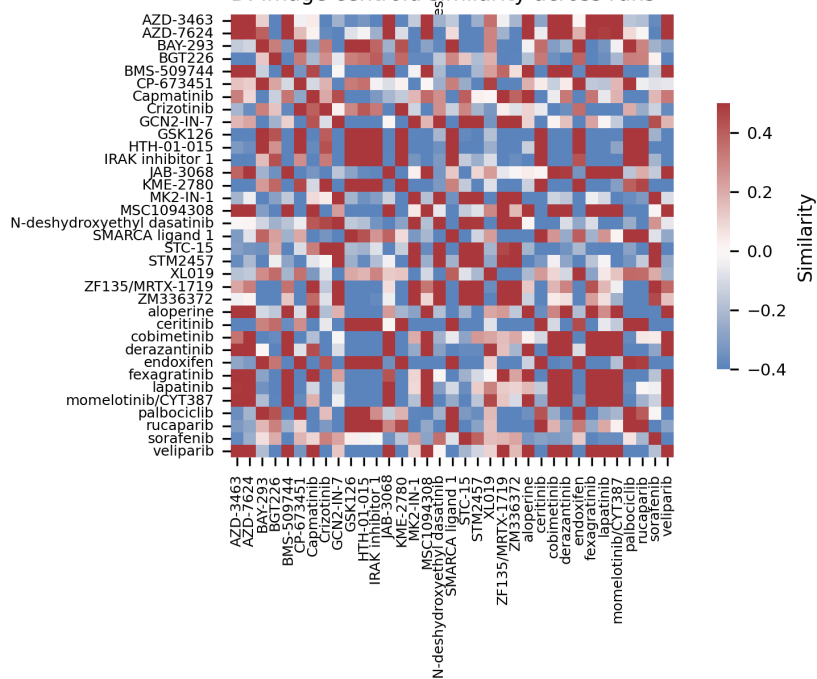


Figure 1. Same-well reproducibility and cross-run centroid transfer in the 35-compound control panel. Both modalities are highly reproducible within run, and both transfer well above chance from ACT010 to ACT011. The calibrated image latent reaches 85.7% top-1 cross-run retrieval and 100% top-3.

Image and RNA preserve a shared compound geometry

Beyond exact label recovery, an additional question for a screening platform is whether the two views rank compounds consistently: if RNA places compound A nearer to compound B than to compound C, does the image latent agree?

Within each run, agreement was substantial. Spearman correlations between pairwise compound-distance matrices were 0.718 in ACT010 and 0.607 in ACT011. The strongest perturbagens recurred across runs: cross-run effect-magnitude rankings correlated at Spearman 0.799 in image space and 0.662 in RNA space. Several compounds (GCN2-IN-7, KME-2780, BGT226, ZF135 / MRTX-1719, CP-673451) appeared near the top of both modalities in at least one run.

Image-derived neighborhoods are therefore biologically aligned with RNA-derived neighborhoods to a degree that supports compound prioritization on either branch, and the largest phenotypic responses generalize across independent experiments. That is the property required for either readout to be used interchangeably during triage.

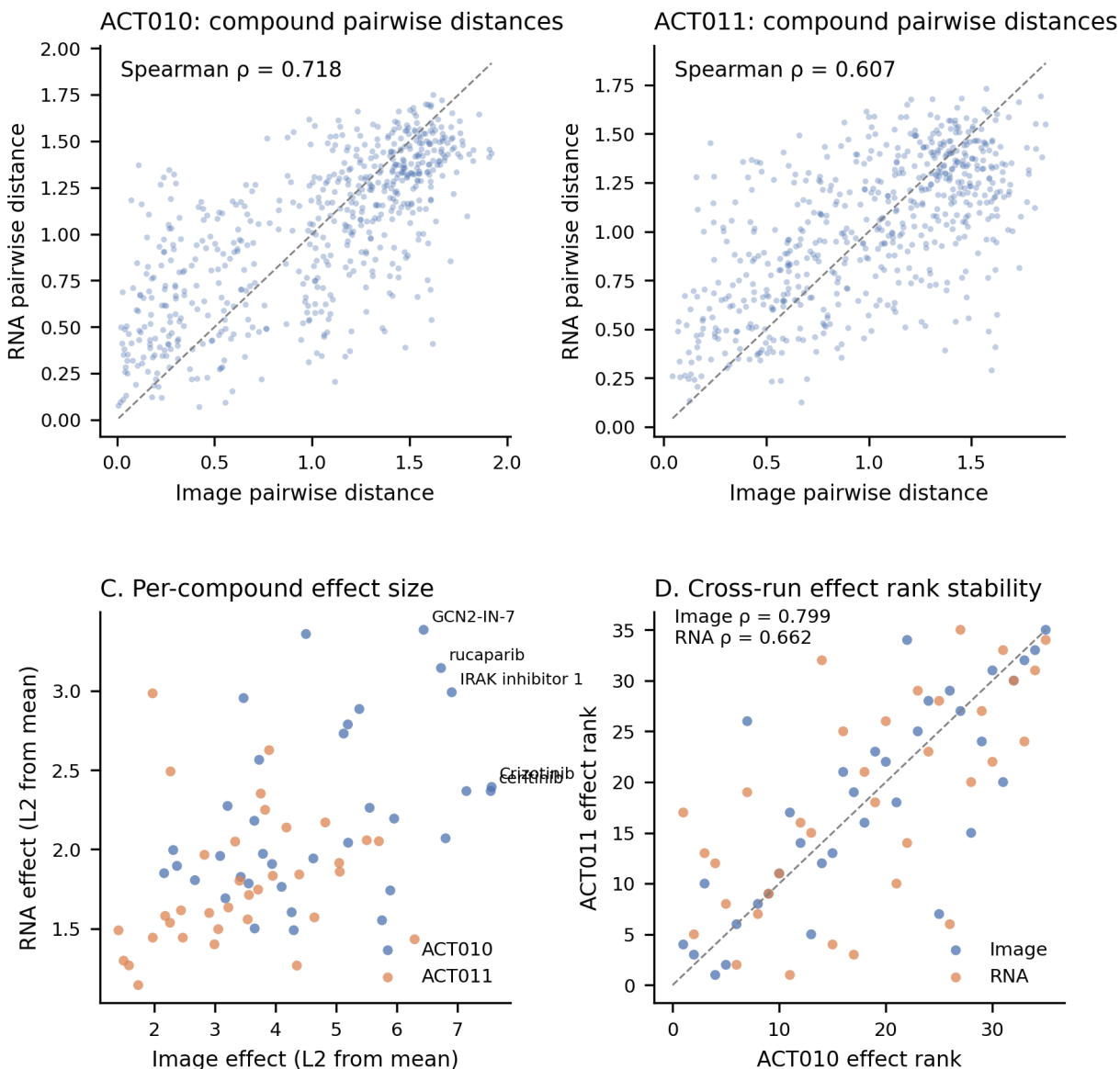


Figure 2. Image and RNA preserve a shared compound geometry within run, and effect-size rankings are stable across runs. Pairwise compound-distance correlations are 0.718 (ACT010) and 0.607 (ACT011); cross-run per-compound effect magnitudes correlate at Spearman 0.799 (image) and 0.662 (RNA).

Same-well multimodality is additive on held-out compound classification

Held-out compound classification is the most operationally relevant benchmark in the dataset. Each matched well was reduced to one calibrated image vector and one RNA vector. A linear discriminant model was trained either on the image vector alone, the RNA vector alone, or the raw concatenation of the two.

Merged image-plus-RNA features outperformed RNA alone in every cut: 55.5% balanced accuracy

versus 41.4% pooled, 60.8% vs 47.3% within ACT010, and 45.1% vs 30.5% within ACT011 (image alone reached 29.2% pooled). Overall accuracy followed the same pattern, rising from 42.5% to 54.9% pooled, from 48.3% to 60.2% in ACT010, and from 31.6% to 45.5% in ACT011.

The gain was broad rather than concentrated in a few easy classes. Merged features improved per-compound recall for 35 of 35 compounds in the pooled analysis, 33 of 35 in ACT010, and 30 of 35 in ACT011. ACT010 confusion matrices show the same pattern: RNA performs well on its own, and adding the calibrated image latent moves probability mass onto correct diagonal entries and resolves several near-confusions.

Linear discriminant analysis on raw-concatenated features was used in preference to a deep classifier because its decision boundary is a linear function of the input features; any reported gain is therefore traceable to a specific axis of the input space rather than to model flexibility.

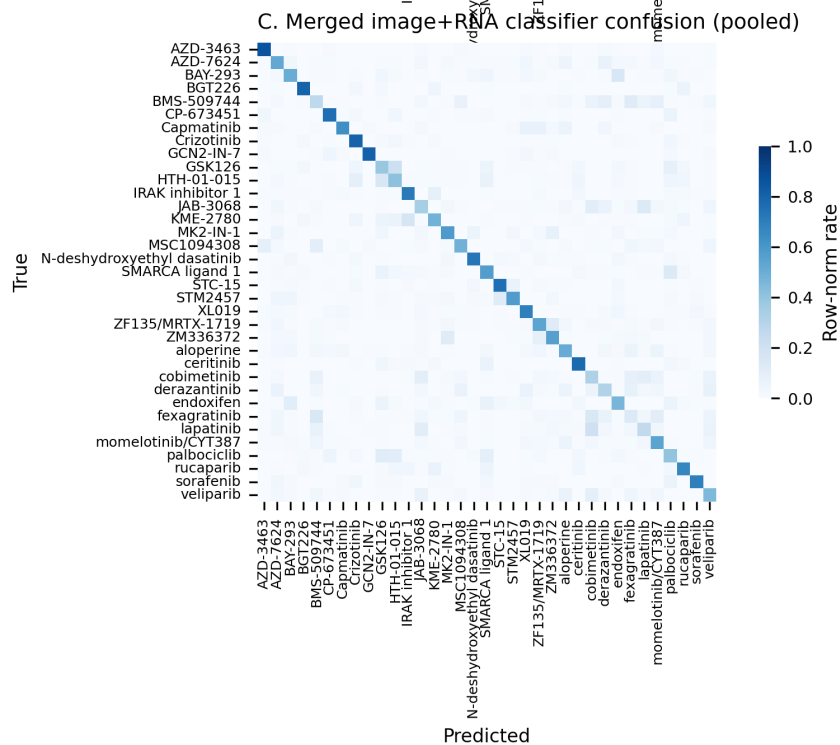
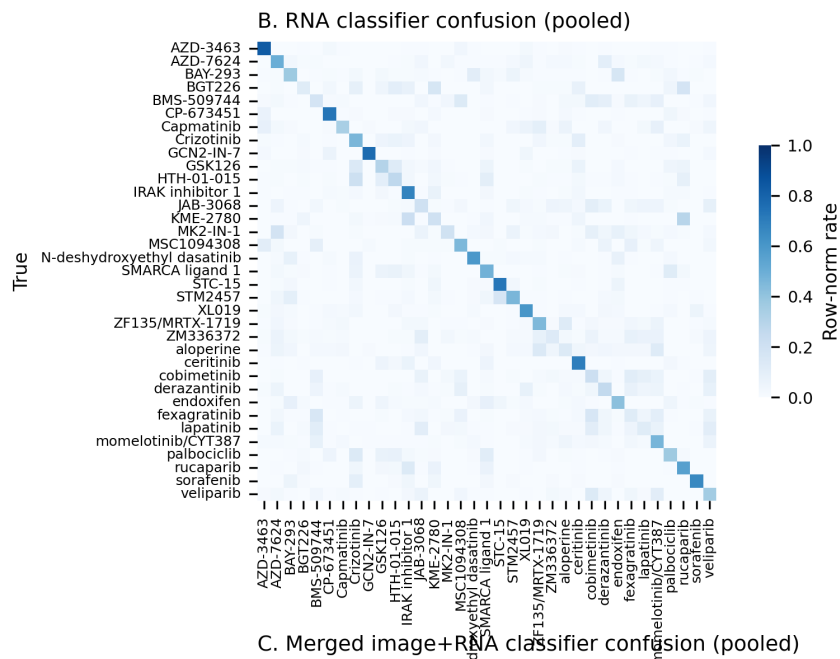
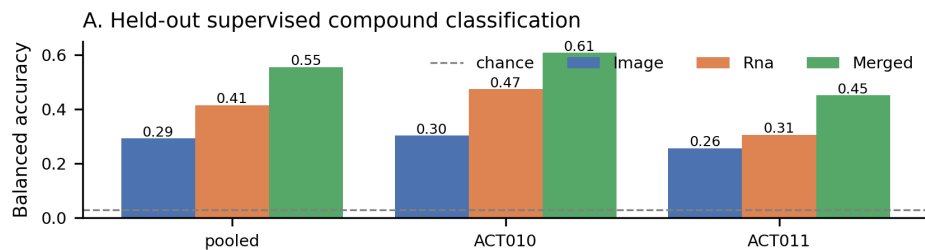


Figure 3. Same-well multimodal classification on held-out wells. Merged image-plus-RNA features outperform RNA alone in pooled and per-run analyses, and per-compound confusion matrices show the gain is from reduced class confusion across most compounds rather than from a few easy classes.

Same-well structure persists beyond the supervised task

A second question is whether same-well pairing carries useful signal when class labels are unavailable at retrieval time, the regime in which hits are typically triaged.

In the harder held-out centroid-retrieval task, the calibrated image latent reached 25.1% / 21.7% top-1 across the two runs, RNA reached 44.8% / 24.3%, and a naive image-plus-RNA concatenation reached 49.1% / 33.7%, exceeding RNA alone in both runs. A lightweight CLIP-style same-well contrastive objective on top of the calibrated latent did not improve retrieval further, indicating that calibration had already exposed most of the recoverable compound-identity information.

Exact well-to-well retrieval remained modest in absolute terms, as expected in a low-pass multicellular assay where 5 to 10 cells per well leave real biological heterogeneity within each pseudo-replicate. The residual ambiguity is structured rather than random. Image and RNA agreed on within-compound well structure above random expectation in 34 of 35 compounds in both runs (cross-modal distance correlation), in 26 of 35 (ACT010) and 29 of 35 (ACT011) by nearest-neighbor overlap, and in 28 of 35 / 27 of 35 by same-compound CLIP retrieval. The pattern is consistent with same-well pairing capturing real within-compound heterogeneity (subpopulation structure, variable response states) rather than only technical scatter.

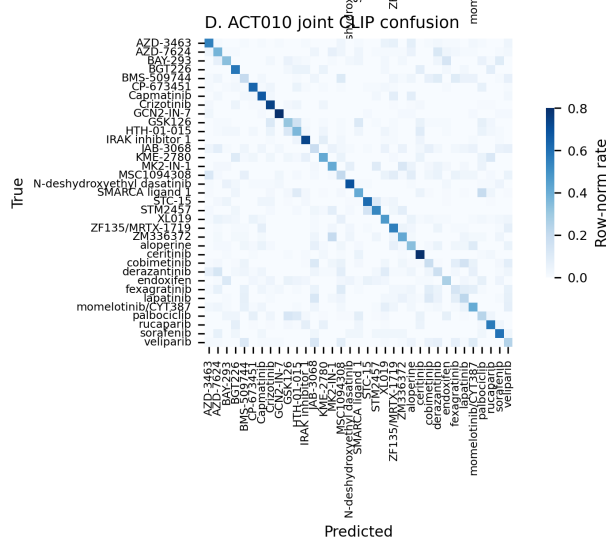
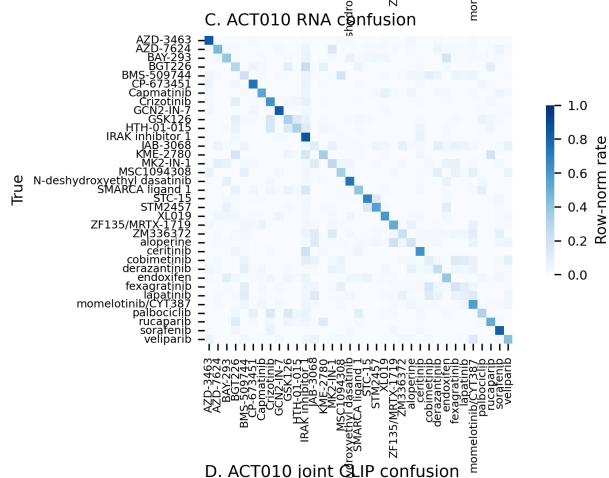
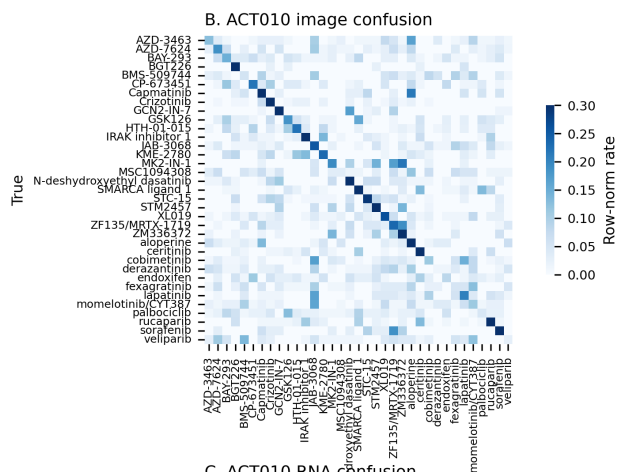
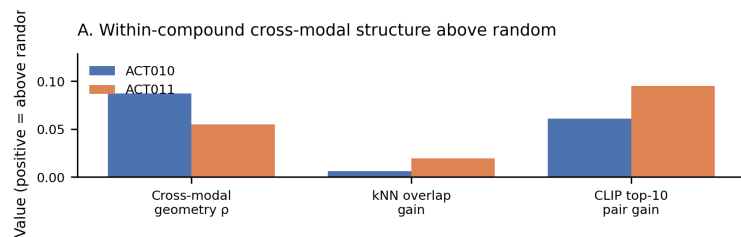


Figure 4. Same-well structure beyond supervised classification. Within-compound cross-modal agreement exceeds random expectation across nearly all compounds, and held-out confusion structure for image, RNA, and joint embeddings shows that residual errors are biologically structured rather than uniform.

Low-pass gene programs remain portable across runs

A screening assay becomes more useful when the underlying gene-level signal remains interpretable. Raw counts from matched control wells were aggregated into per-compound pseudobulk signatures, converted to log-CPM, and centered by subtracting each run’s experiment-wide mean. Pseudobulk plus centering was used in preference to per-cell differential-expression modeling because each well averages 5 to 10 cells with low per-well RNA depth; pseudobulk is the unit at which the signal is robust, and centering removes experiment-level offsets without imposing a normalization across two physically separate runs.

Centered signatures retrieved the correct compound across runs at 48.6% top-1 and 68.6% top-3. Representative compounds carried coherent gene programs: palbociclib (ARC, RASD1, CDKN1C, EGR1), BAY-293 (MT1X, MT1F, MT2A, SLC30A1), rucaparib (ITPRIP, SLC37A4, NACC2, COL4A1), crizotinib (SORCS2, ZNF34, AC253572.2), GSK126 (DUSP10, SPATA6L, CCN2). Validation of each gene as a mechanism-of-action marker in HEK293 is a separate question. The platform-level claim is that ActiveSeq preserves compound-specific transcriptional structure at the gene level even at present low-pass depth, supporting the use of transcriptomics for hit de-risking and phenotype clustering after primary screening.

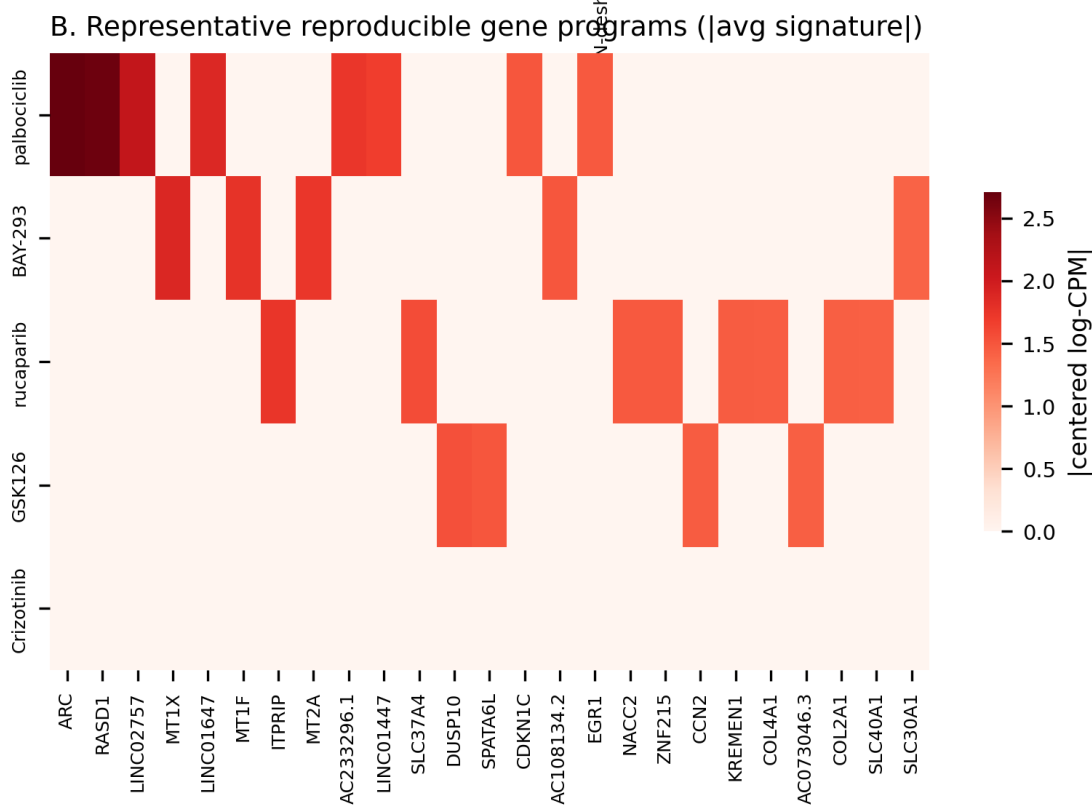
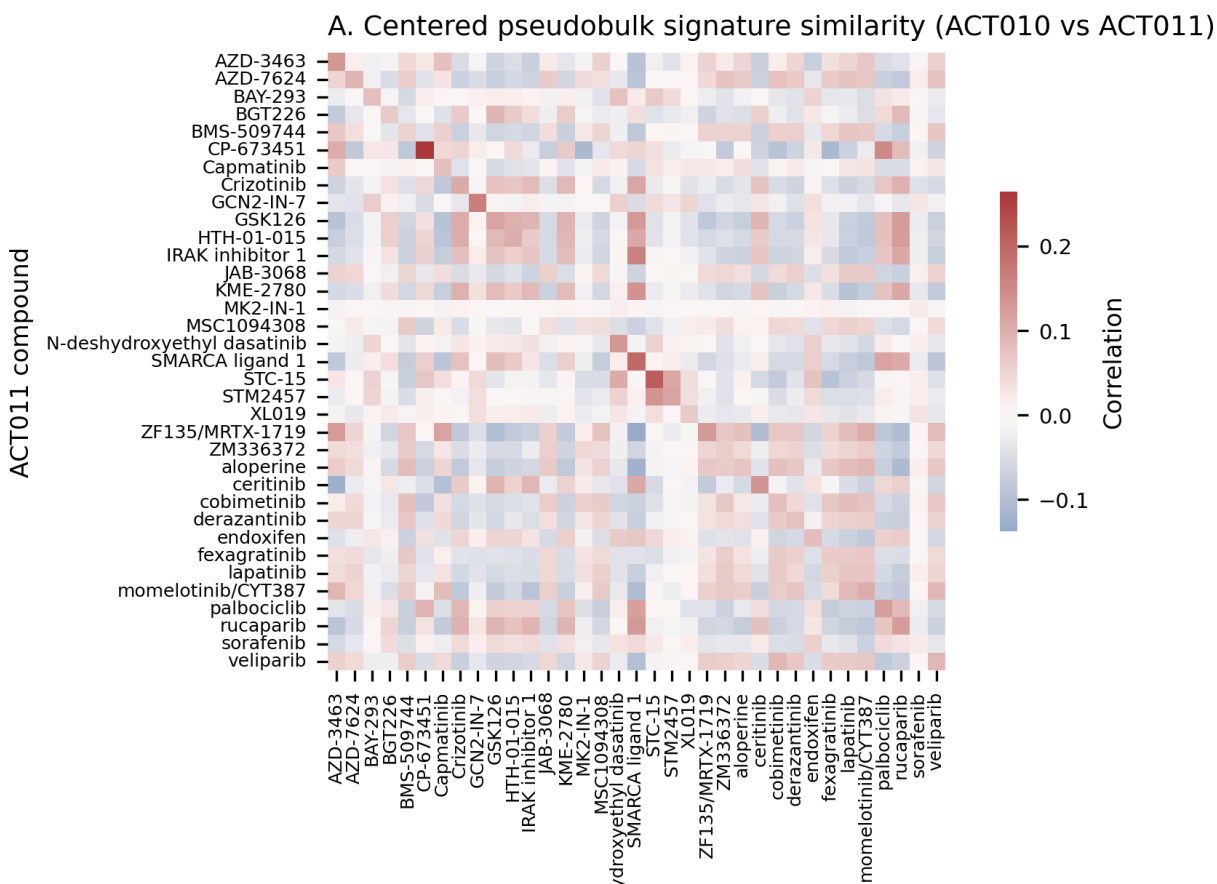


Figure 5. Centered pseudobulk gene signatures preserve compound-specific transcriptional programs across runs and recover the correct compound at 48.6% top-1 and 68.6% top-3 retrieval, with coherent gene programs visible for representative compounds.

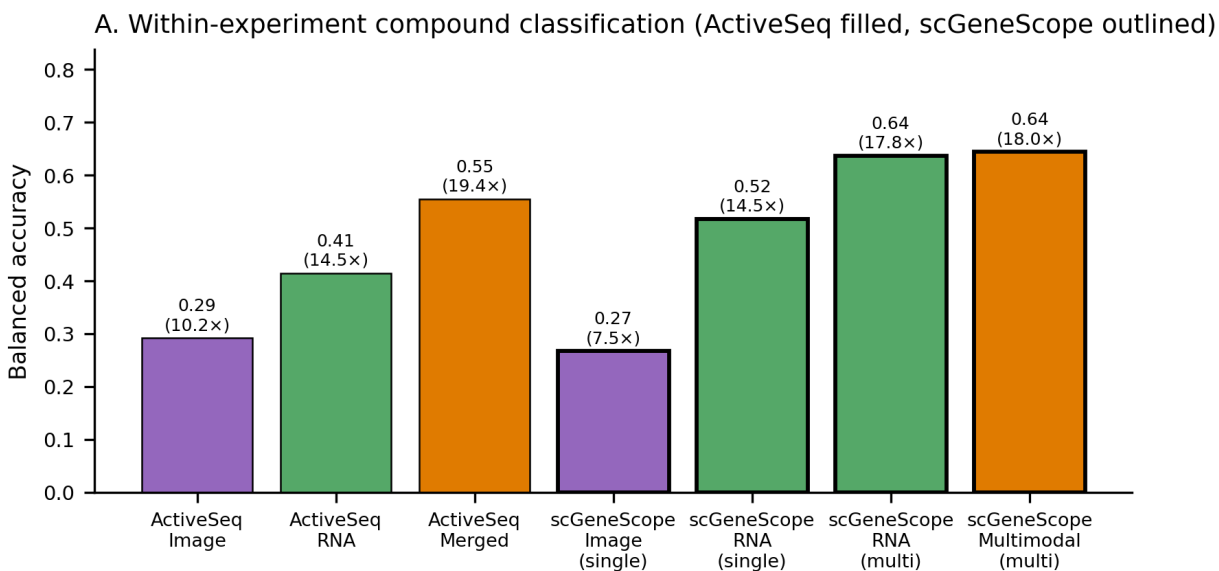
Position relative to scGeneScope and the broader phenotypic-screening landscape

Three reference points anchor the platform comparison. The first is Cell Painting and the imaging-only ML platforms built on it (Recursion’s RxRx public datasets, Insitro’s morphology programs, the Broad JUMP consortium). These systems lead on raw image throughput by an order of magnitude or more and have produced reusable embedding spaces, at the cost of biological readout depth. The second is treatment-matched multimodal benchmarks, of which scGeneScope is the strongest current public example: approximately 716,767 single-cell images and 627,704 single-cell RNA profiles across 28 treatments in U2-OS cells, paired at the level of replicate populations rather than the same well [4]. The third is the perturbational scRNA-seq line (Perturb-seq, Tahoe-100M, scPerturb), which provides the deepest available cell-state readout but does not include matched imaging.

Z-Screen / ActiveSeq combines all three properties: the imaging pass operates at Cell Painting scale, the sequencing pass is run inside the same micro-well as the imaging, and chemistry is delivered through one-bead-one-compound combinatorial libraries so every well carries full chemistry provenance. The cost of that design is dataset size: the present pilot comprises 11,511 control-panel matched wells across two experiments at 5 to 10 cells per well, equivalent to approximately 60,000 to 110,000 cells in the control panel, or roughly seven to twelve times fewer total cells profiled than scGeneScope, on a simpler imaging modality. Z-Screen reads each well as a pseudobulk rather than at single-cell resolution, so the unit of independent observation is the well, not the cell.

ActiveSeq trails the strongest scGeneScope multimodal reference on raw within-experiment balanced accuracy, consistent with the scale and imaging-richness gap. The more informative comparison is fold over random-chance performance, which normalizes for the different label spaces (35 versus 28 classes). On that scale, pooled ActiveSeq RNA reaches 14.5 times chance, identical to the scGeneScope single-profile RNA value of 14.5 times chance. Pooled ActiveSeq image reaches 10.2 times chance, above the scGeneScope single-profile image value of 7.5 times chance. Pooled ActiveSeq merged features reach 19.4 times chance, slightly above the scGeneScope multimodal multiprofile reference at 18.0 times chance, and the strongest ActiveSeq run (ACT010) reaches 21.3 times chance for merged features.

Once task difficulty is normalized, a small same-well pilot with approximately an order of magnitude fewer total cells profiled and a simpler imaging modality reaches the same range as the strongest public multimodal benchmark, while preserving a tighter same-well pairing unit and exposing recoverable image-side value through a calibrated latent. The pilot uses two experiments and one cell line on a platform designed to scale by orders of magnitude beyond the present footprint, so the result is directional and motivates collecting substantially more matched data rather than serving as a final benchmark claim.



B. Context for interpreting the gap

Aspect	ActiveSeq (Z-Screen)	scGeneScope
Single-cell scale	~60-110k cells (5-10 × 11.5k wells)	~717k images, ~628k RNA
Label space	35 compounds	28 treatments
Pairing	Same matched micro-well	Treatment-matched replicate
Imaging	Brightfield + calibrated latent	5-ch Cell Painting + ViT
Pooled image	29.2% (10.2× chance)	26.7% single (7.5×)
Pooled RNA	41.4% (14.5× chance)	51.7% single (14.5×)
Pooled multimodal	55.5% (19.4×)	64.4% multi (18.0×)
Best ActiveSeq run	ACT010 merged 60.8% (21.3×)	n/a
Takeaway	Smaller pilot, tighter pairing; competitive	Larger public benchmark on raw accuracy

Figure 6. Within-experiment compound classification for ActiveSeq versus the published scGeneScope reference, annotated with fold over random-chance performance, plus a context table summarizing the design differences that make the two benchmarks complementary rather than directly interchangeable.

ZEL026 dummy-bead wells perform as a negative-control QC

The ZEL026 dummy-bead wells contain no compound and were retained as a negative-control assay-chemistry stress test rather than as a discovery result. They behaved as expected. Median UMIs dropped from 5,167 to 342 in ACT010 and from 4,750 to 293 in ACT011; detected genes dropped from 2,826 to 301 and from 2,463 to 248 respectively. The collapse is operationally useful: the failure mode is visible in the readout itself and does not contaminate the 35-compound biological analyses. The QC distribution is reported in Supplementary Figure 1.

Discussion

The dataset is small by design, and the questions it answers are foundational. ActiveSeq produces stable compound-level multimodal structure inside the Z-Screen microchip. The RNA branch is strong within and across runs. After task-aware calibration, the image branch carries enough aligned signal to add value at present scale rather than requiring a much larger collection effort. Same-well pairing produces an additive gain on the most practical benchmark evaluated here: a merged representation outperforms RNA alone on held-out compound classification, and the gain extends to all 35 compounds in pooled analysis. Adding matched transcriptomics to imaging in a 50,000-well one-bead-one-compound screen therefore produces measurable operational value rather than only a richer descriptive readout.

The competitive context is straightforward. Imaging-only ML platforms compete on imaging scale and library coverage; transcriptomic platforms compete on cell-state readout depth per compound. Both axes are necessary to address mechanism questions, and neither is sufficient on its own. Z-Screen is engineered to do both inside the same micro-well, a design point not occupied by current public datasets. The pilot reported here is small by intent, and the result is that the design point produces a broad gain across compounds rather than one concentrated in a few easy classes.

For a discovery team, ActiveSeq turns the Z-Screen microchip into a same-well multimodal phenotype engine: imaging provides breadth, transcriptomics provides mechanism, and the two are coupled at the level of the same multicellular colony rather than at treatment-matched populations. That coupling is the property most likely to scale into a primary-screening readout, and it is not provided by current public benchmarks.

The natural next experiments follow directly: enlarge the control panel, broaden cell-line coverage, continue improving imaging consistency to strengthen the morphology contribution, and continue collecting matched same-well pairs at the throughput the chip is designed to support. Each investment compounds a signal already visible in the present dataset.

Methods

Dataset

The analysis uses the canonical Z-Screen ActiveSeq workspace under `data/ZScreen_Canonical_Dataset/`. Each ActiveSeq well is annotated with an experiment id (ACT010 or ACT011), a global well

id, a compound assignment, an image embedding, and an RNA latent. Gene-level pseudobulk counts come from the matched H5AD object that backs the same-well RNA pipeline, repaired as documented in `DATA_REPAIR_REPORT.md`. The scGeneScope comparison uses published benchmark values transcribed from OpenReview Table 1 into `scgenescope_reference.csv`.

Per-well image embedding calibration

The original per-well image embedding was produced by a frozen pretrained vision model on raw imaging tiles. That representation captures generic visual structure but does not necessarily align compound identity with the principal directions of variance: experiment-of-origin, plate position, and microscopy artifacts can dominate compound signal in the raw geometry. Direct held-out compound classification on that embedding therefore tests the representation as much as the biology.

A 32-dimensional control-panel-calibrated latent was learned from the original embedding by a lightweight domain-adversarial network trained on the matched control panel. The objective rewards compound discrimination and penalizes experiment-of-origin discrimination, which pushes compound information onto the principal directions of variance without retraining a vision model from scratch. Calibration was learned on the same control-panel domain studied here, so the calibrated-latent results should be read as an internal representation refinement; broader-library and additional-cell-line generalization is the next test.

Matched-well preprocessing

Imaging detections were aggregated by `experiment_id` and `gwid` so each matched well contributed one image vector. Wells with an assigned control name were treated as the 35-compound panel; wells without an assigned control name were the ZEL026 dummy-bead wells used for QC.

Reproducibility and transfer analyses

Within-run reproducibility was measured by repeated split-half pseudo-replication within each compound (cosine similarity between pseudo-replicate centroids after standardization within each matrix). Cross-run transfer used compound centroids learned in ACT010 and queried in ACT011.

Cross-modal geometry

For each run, pairwise distances among compound centroids were computed separately in image and RNA space. Agreement between distance matrices was summarized with Spearman correlation over upper-triangular entries, which does not assume the two scales are matched. Cross-run effect-size stability used Spearman correlation over per-compound effect magnitudes.

Supervised multimodal classification

Held-out classification used a 50:50 compound-stratified split, either pooled across runs or fit within each run. Linear discriminant analysis (eigen solver, automatic shrinkage) was trained on calibrated image features alone, RNA features alone, or their raw concatenation. LDA was used in preference

to a deep classifier so that any reported gain is traceable to a linear function of the input features. Performance was summarized with overall accuracy, balanced accuracy, and per-compound recall.

Same-well representation learning

A lightweight CLIP-style dual encoder with a symmetric contrastive objective was trained on same-well pairs from the training split, evaluated by nearest-centroid compound retrieval and by exact held-out pair retrieval. The contrastive objective did not exceed naive concatenation in this dataset, indicating that most of the recoverable compound-identity signal had already been exposed by the calibration step.

Gene-level pseudobulk analysis

Raw counts were summed per experiment and control compound, converted to log-CPM, and centered by subtracting the experiment-wide mean signature. Cross-run retrieval used correlation similarity between centered signatures. Per-cell differential expression was not performed because each well averages 5 to 10 cells; pseudobulk plus centering is the unit at which the signal is robust at present depth.

External benchmark context

Within-run supervised metrics from `supervised_multimodal_metrics.csv` were compared to scGeneScope within-experiment values transcribed from OpenReview Table 1 into `scgenescope_reference.csv`. Because the label spaces differ (35 vs 28), the comparison reports both raw balanced accuracy and fold over random-chance performance.

Limitations

The pilot is intentionally narrow: one cell line (HEK293), 35 control compounds, and two matched experiments. The calibrated image latent was learned on the same control panel studied here; broader libraries and additional cell lines are needed to evaluate calibration generalization. Exact same-well cross-modal retrieval remains modest in absolute terms, as expected in a low-pass multicellular assay. Gene-level signatures are summarized through pseudobulk aggregation rather than full differential-expression modeling with biological replicates. These caveats accompany rather than offset the central conclusion: same-well matched imaging and transcriptomics in the Z-Screen microchip support reproducible compound-resolved multimodal structure, and same-well pairing produces an additive gain on held-out classification.

Data availability

All data tables, derived latent representations, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper1/` and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper1/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

References

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4. Dapello J, Nassar M, Eksi R, et al. scGeneScope: A treatment-matched single-cell imaging and transcriptomics dataset and benchmark for treatment response modeling. *NeurIPS 2025 Datasets and Benchmarks.* OpenReview. <https://openreview.net/pdf/f7d541dae38bcf88a79789a4c6440aadfec123c7>